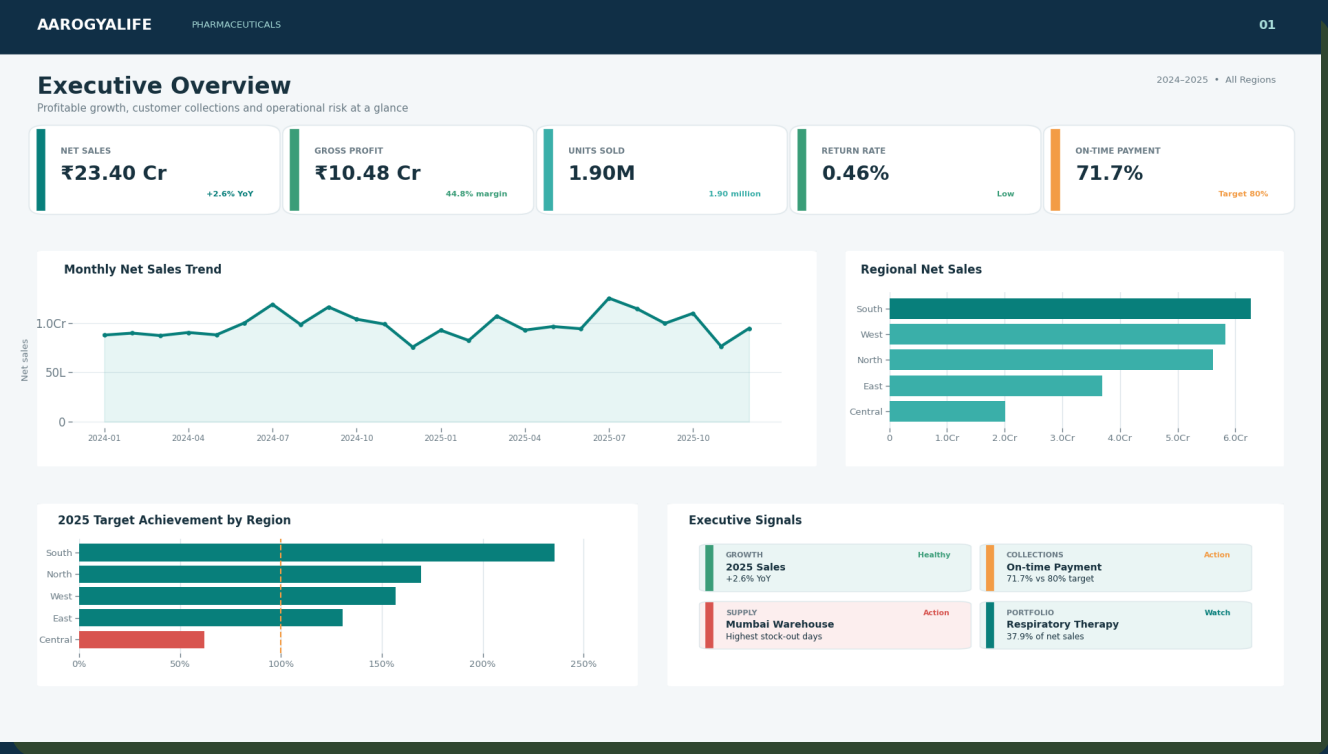


Pharmaceutical growth without hidden risk

Business Problem and Required Deliverables

AarogyaLife Pharmaceuticals needed a connected analytical view of commercial performance, inventory availability, payment discipline and safety-reporting workload.



BUSINESS PROBLEM

Revenue growth alone could not explain overall business health

Leadership had data across separate operational tables, but lacked one trusted view to understand where growth was profitable, serviceable and collectible.

01

Commercial blind spots

Which regions, therapies, products and channels were creating profitable growth?

02

Availability pressure

Were stock-outs, reorder gaps and damaged stock limiting high-demand products?

03

Cash collection risk

Which channels and distributors were paying beyond agreed payment terms?

04

Safety workload

Which reports required faster operational review and follow-up?

CENTRAL BUSINESS QUESTION

Where can the company grow without increasing returns, payment delays, stock-outs or unresolved safety-reporting workload?

STAKEHOLDERS

Executive leadership | Commercial team | Supply chain | Finance and collections | Safety operations

Scope note: the dataset is synthetic and contains no personally identifiable patient information. Safety-reporting counts support workload prioritisation only; they do not establish incidence or causality.

REQUIRED DELIVERABLES

A complete analytical solution from raw CSV files to business decisions

The project must be reproducible, validated and easy for business stakeholders to explore.

1

Python data preparation

Profile six CSV files; standardise dates and categories; check duplicates, missing keys, invalid values and calculation consistency.

2

MySQL analytical layer

Create related tables with primary and foreign keys; validate row counts; build reusable SQL queries and views for commercial, inventory, collections and safety analysis.

3

Power BI semantic model

Create a star schema, Date table, relationships and reusable DAX measures for sales, margin, returns, targets, payments, inventory and reporting workload.

4

Four-page dashboard

Executive Overview; Commercial Performance; Supply Chain and Inventory; Safety Monitoring - with consistent filters, KPIs and decision-focused insights.

DEFINITION OF DONE

Numbers reconcile across Python, SQL and Power BI; dashboards contain no clipped visuals; insights lead to clear actions for growth, collections, inventory and reporting workflow.